IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

DIGITAL SIGNAL PROCESSING AND DIGITAL FILTERS

Tuesday 9 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper. Answer Question 1 and any TWO other questions.

Question 1 is worth 40% of the marks and other questions are worth 30%

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : P.T. Stathaki

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Question 1 [40 Marks]

a) Real-world signals are **[2 Marks]**

(a) analog in amplitude,

(b) quantized in amplitude,

(c) vary continuously in time,

(d) are discrete in time.

(Multiple answers are possible.)

b) How is a discrete signal scaled so that the sampling frequency satisfies $f_s = 1$? If the original analog signal has bandwidth Ω rad/s the scaled samples represent samples from an analog signal of what bandwidth? **[2 Marks]**

c) Prove that discrete convolution is associative and commutative. **[2 Marks]**

d) Let $H(e^{\jmath\omega})$ be a zero-phase filter of odd length $M+1$, symmetric around 0, i.e., $h[-n] = h[n]$. Furthermore, assume that $h[n]$ is real valued. The weighted error is $e(\omega) = s(\omega)\left(H(e^{\jmath\omega} - d(\omega)\right)$, where $d(\omega)$ is the target filter satisfying

$$d(\omega) = \begin{cases} 1, & 0 \le \omega \le \omega_1, \\ 0, & \omega_2 \le \omega \le \pi. \end{cases} \tag{1}$$

Choose $s(\omega)$ to control the error variation with ω. **[2 Marks]**

e) Let the system $G(z)$ be defined as

$$G(z) = \frac{8 - 2z^{-1}}{4 - (2+\alpha)z^{-1} + 3z^{-2}}. \tag{2}$$

Find $\alpha \in \mathbb{R}$ such that $G(z)$ has two real poles. **[1 Marks]**

f) The input/output relation for a signal processing system is described by,

$$2u\,[n] \longrightarrow \boxed{h\,[n]} \longrightarrow \left(\frac{2^2}{2^n} - 3\left(-\frac{3}{4}\right)^n\right) u\,[n]$$

where $u[n]$ is the usual unit-step sequence.

Find the impulse response of this system. **[3 Marks]**

g) Suppose that $s\,(t)$ is a bandlimited signal with maximum frequency of $\Omega = \pi$ (rad/sec).

i) What is the largest sampling period T that allows for reconstruction of $s(t)$ from $s[k] = s(kT)$? **[1 Marks]**

ii) Assume that we are given samples $s[k]$. Derive an algorithm to obtain τ-delayed samples,

$$s_\tau[k] = s(kT - \tau)$$

where $0 < \tau < T$. **[3 Marks]**

h) A filter is defined by its complex-valued pole $\lambda_0 \in \mathbb{C}$,

$$\frac{B(z, \lambda_0)}{A(z, \lambda_0)} = \frac{z^{-1} - \lambda_0^*}{1 - \lambda_0 z^{-1}}$$

where λ_0^* is the complex-conjugate of λ_0.

i) Show that $B(z, \lambda_0)/A(z, \lambda_0)$ defines an all-pass filter. **[1 Marks]**

ii) Show that the cascaded representation,

$$\text{Input} \longrightarrow \underbrace{\boxed{\frac{B(z, \lambda_0)}{A(z, \lambda_0)}} \longrightarrow \boxed{\frac{B(z, \lambda_0^*)}{A(z, \lambda_0^*)}}}_{H(z)} \longrightarrow \text{Output}$$

allows for an implementation using real-valued coefficients only. Letting,

$$H(z) = \frac{\sum_{m=0}^{M-1} b_m z^{-m}}{\sum_{m=0}^{M-1} a_m z^{-m}}$$

identify M, a_m and b_m. **[3 Marks]**

iii) For an input sequence $x[k]$, write the filtered output $y[k]$ in terms of the real-valued coefficients of filter $H(z)$ above.

[1 Marks]

i) **Hilbert Relations.** Let us define a sequence,

$$s[k] = \begin{cases} 0 & k = 0 \\ \frac{k}{|k|} & k \neq 0 \end{cases}.$$

i) By letting $S(z) = \sum_k s[k] z^{-k}$, show that $\left(1 - z^{-1}\right) S(z) = 1 + z^{-1}$. **[2 Marks]**

ii) Evaluate $S\left(e^{\jmath\omega}\right)$. **[1 Marks]**

iii) Let $\overline{h}(t) = h(-t)$.

Based on this, let us define the following,

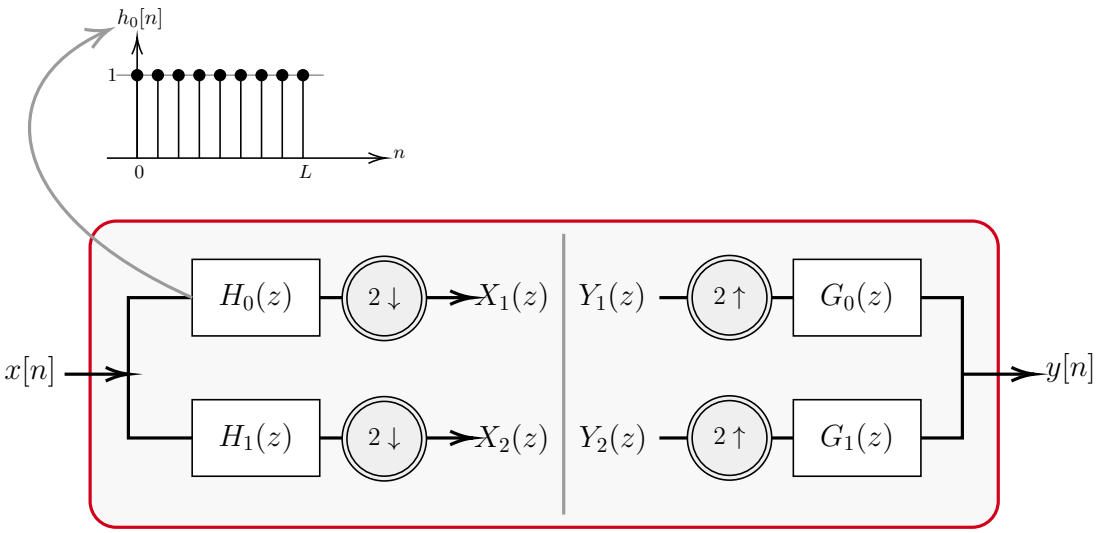
$$h_E[n] = \frac{1}{2}\left(h[n] + \overline{h}[n]\right) \quad \text{and} \quad h_O[n] = \frac{1}{2}\left(h[n] - \overline{h}[n]\right).$$

leading to Fourier relations, $\Re\left(H\left(e^{\jmath\omega}\right)\right) = H_E\left(e^{\jmath\omega}\right)$ and $\Im\left(H\left(e^{\jmath\omega}\right)\right) = -\jmath H_O\left(e^{\jmath\omega}\right)$. Show that whenever $h[n]$ is a causal sequence, it follows that real and imaginary parts of the spectrum of $h[n]$ are related to each other. In particular,

$$\Im\left(H\left(e^{\jmath\omega}\right)\right) = -\Re\left(H\left(e^{\jmath\omega}\right)\right) \odot \cot\left(\frac{\omega}{2}\right)$$

where $\odot$ denotes circular convolution. **[2 Marks]**

j) The input-output relationship for a particular signal processing system is described by,



where the filter $H_0(z) = \sum_{n=0}^{L} z^{-n}$.

The input signal $x[n]$ undergoes filtering and decimation or downsampling (**2** $\downarrow$). This is the signal analysis part. There on, given sequences or transfer functions, the synthesis part reconstructs the original signal by upsampling (**2** $\uparrow$) followed by filtering. In the above H_0 and H_1 are low-pass and high-pass filters, respectively.

i) Identify outputs $X_1(z)$ and $X_2(z)$ given $x[n]$, $H_0(z)$ and $H_1(z)$. **[2 Marks]**

ii) How is $y[n]$ related to inputs transfer functions $Y_1(z)$ and $Y_2(z)$? **[2 Marks]**

iii) Suppose that one system's output becomes another system's input, that is,

$$Y_1(z) = X_1(z) \quad \text{and} \quad Y_2(z) = X_2(z).$$

What is the frequency response of $y[n]$? **[2 Marks]**

Under what conditions (in frequency domain), $y[n] = x[n]$? **[2 Marks]**

Let $h[n]$ be the impulse response of some filter $H(z)$. Furthermore, let $H_0(\omega) = H(\omega)$ and $H_1(\omega) = H(\omega - \pi)$. Identify the impulse response of $h_1[n]$, $g_0[n]$ and $g_1[n]$ in terms of $h[n]$. **[2 Marks]**

k) Let

$$X[k] = \sum_{n=0}^{N-1} x[n] \cos\left(\frac{2\pi}{4N}(2n+1)k\right)$$

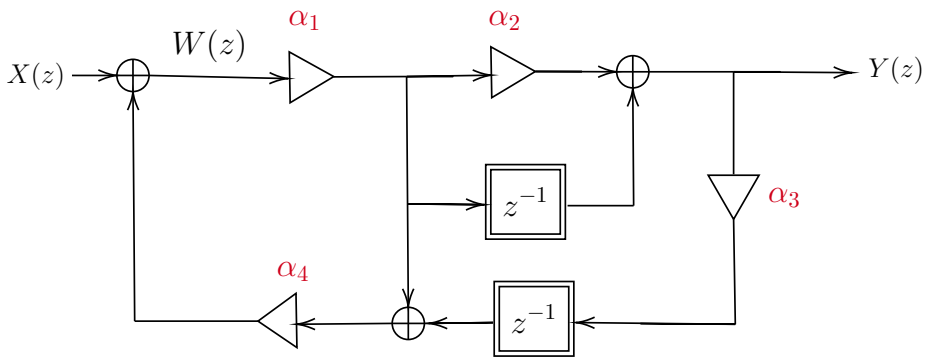
define the Discrete Cosine Transform of some sequence $x[n]$. Prove the anti-periodic and periodic property of $X[n]$, that is,

$$\underbrace{X[k+2N] + X[k] = 0}_{\text{Anti-Periodic}} \quad \text{and} \quad \underbrace{X[k+4N] - X[k] = 0}_{\text{Periodic}}.$$

[4 Marks]

Question 2 [30 Marks]

a) Let $H(z) = \frac{Y(z)}{X(z)}$ be a system represented in the diagram below



i) Derive the expression of $W(z)$ in terms of $X(z)$. **[3 Marks]**

ii) Derive the expression of the transfer function $H(z) = \frac{Y(z)}{X(z)}$. **[4 Marks]**

iii) What are the conditions that ensure that $H(z)$ is BIBO-stable? **[1 Mark]**

iv) Sketch the pole-zero plot for $\alpha_1 = \alpha_3 = 1, \alpha_2 = 2, \alpha_4 = 4$. **[1 Mark]**

b) *i*) Let α be a complex-valued constant, compute the z-transform of a causal sequence $x[n] = e^{\alpha^2 n}$ and specify its region of convergence. **[2 Marks]**

ii) Suppose that $G_1(z) = 1 - \alpha z^{-1}$ and $G_2(z) = 1 - \beta z^{-1}$ where the constants α and β may be complex. If $\alpha\beta^* = 1$, show that $|G_1(e^{\jmath\omega})| = p|G_2(e^{\jmath\omega})|$ for all ω and determine the constant p. **[3 Marks]**

iii) Let $H_1(z) = 3z^{-2} - 16z^{-1} - 12$. Find the expression of $H_2(z)$ such that $|H_1(e^{\jmath\omega})| = |H_2(e^{\jmath\omega})|$ for all ω and that all the zeros of $H_2(z)$ lie inside the unit circle. **[3 Marks]**

iv) The impulse response, $h[n]$, of a linear time-invariant system satisfies $\sum_{n=-\infty}^{\infty} |h[n]| = S < \infty$. Show that the system is BIBO-stable and also that $H(z)$ converges for $|z| = 1$. **[3 Marks]**

c) A filter, with input $x[n]$ and output $y[n]$, is defined by the difference equation

$$y[n] = \frac{2-\alpha}{1+\alpha} y[n-1] + \frac{2\alpha - 1}{1+\alpha} x[n]$$

where $0 < \alpha < 1$ is a real constant.

i) Determine the system function of the filter, $H(z)$, and the impulse response, $h[n]$, for $n = -1, 0, 1$. **[3 Marks]**

ii) State the values of z at which $H(z)$ has a pole or zero. **[2 Marks]**

iii) Determine the frequency at which the filter has a gain of -3dB. **[2 Marks]**

iv) If the sample frequency is f_s, show that, for $n \geq 0$, the impulse response, $h[n]$, is equal to a sampled version of $f(t) = Ae^{-t/\beta}$ and determine the values of the constants A and β. **[3 Marks]**

Question 3 [30/30 Marks]

a) Assume that the filters $P(z^2) = \frac{\alpha+z^{-2}}{1+\alpha z^{-2}}$ and $Q(z^2) = \frac{\beta+z^{-2}}{1+\beta z^{-2}}$ are stable. Let $H(z) = P(z^2) + z^{-1}Q(z^2)$. Determine the values of α and β so that the numerator of $H(z)$ is a scalar multiple of $(1+z^{-1})^5$. **[5 Marks]**

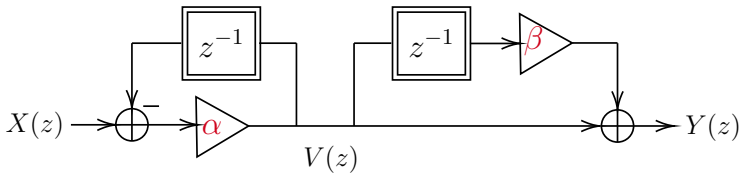
b) The impulse response of an antisymmetric FIR filter, $H(z)$, of order M satisfies the relation $h[n] = h[M-n]$.

i) Show that the magnitude response $|H(e^{\jmath\omega})|$ can be expressed as the absolute value of the sum of N cosine functions where $N = M/2$ if M is even and $N = (M+1)/2$ if M is odd. **[3 Marks]**

ii) Show that $H(e^{\jmath\omega})$ is necessarily zero at $\omega = \pi$ if M is odd. provide an example of a filter for which this is the case. **[2 Marks]**

iii) Derive an expression for the phase response, $\angle H(e^{\jmath\omega})$, and determine the group delay, $\tau_H(e^{\jmath\omega}) = \frac{-d\angle H(e^{\jmath\omega})}{d\omega}$. **[3 Marks]**

c) Let $H(z) = \frac{Y(z)}{X(z)}$ be represented by the diagram below.



i) Derive the expression of $V(z)$ **[3 Marks]**

ii) Derive the expression of the transfer function $H(z)$. **[3 Marks]**

iii) Give conditions for which $H(z)$ realizes a first order all-pass transfer. **[2 Marks]**

iv) What are the conditions that ensure that $H(z)$ is BIBO-stable? **[1 Mark]**

d) Consider the system with the following impulse response

$$h[n] = \left(\frac{1}{\lambda}\right)^{n+2} u[n-K],$$

where $u[n]$ denotes the step function and $\lambda > 1$.

i) Derive the phase of the system. **[3 Marks]**

ii) Derive the expression of the group delay $\tau_H(e^{\jmath\omega})$. **[3 Marks]**

iii) Find λ, K such that $\tau_H(e^{\jmath\pi}) = 3$ and $\tau_H\left(e^{\jmath 0}\right) = 1$. **[2 Marks]**

Question 4 [30 Marks]

a) Let $y_N[k]$ be the output of a system with impulse response $h_N[k]$ satisfying

$$y_N[k] = \sum_{n=1}^{N} \frac{x[k-n]}{n(n+1)}.$$

i) Prove that the system is linear and time-invariant. **[2 Marks]**

ii) Compute the system transfer function $H_N(z)$ corresponding to $h_N[k]$.
For $N = 1$, compute analytically and sketch its magnitude and phase response functions.
[3 Marks]

iii) Compute the step response of $H_N(z)$, denoted $y_N^0[k]$.
Find the steady state $\lim_{k\to\infty} y_N^0[k]$. **[2 Marks]**

iv) Let E_x^N be defined as $E_x^N = \sum_{k=-N}^{N} |x[k]|^2$, such that $E_x = \lim_{N\to\infty} E_x^N$ is the "energy" of a sequence $x[k]$. Calculate the expression of $E_{y_N^0}^N$ as a function of N.
Compute $E_y = \lim_{N\to\infty} E_{y_N^0}^N$. **[3 Marks]**

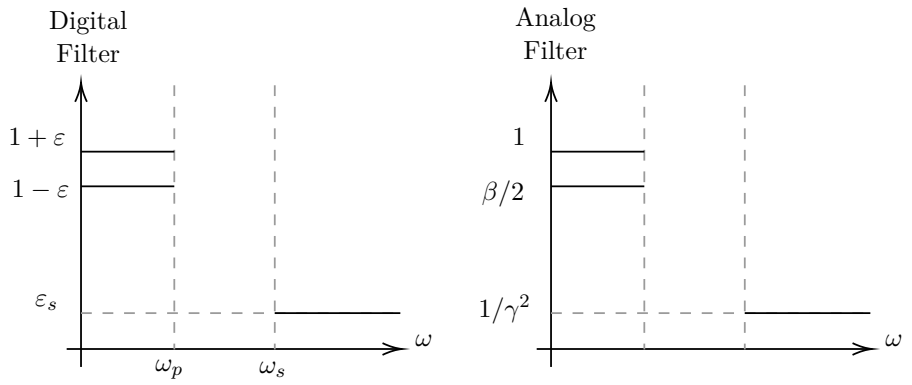
b) Let $H(z) = H_1(z) + z^{-1}H_2(z)$ where $H_1(z)$ and $H_2(z)$ are two filters satisfying

$$H_1(z) = \beta/2 + z^{-1} - z^{-2} - z^{-4}, \tag{3}$$
$$H_2(z) = 2 + (\gamma - 3)z^{-1} - 4z^{-3} + 3z^{-4} - 2z^{-5}. \tag{4}$$

i) Compute β and γ such that $h[n]$ has a linear phase response. **[4 Marks]**

ii) Derive analytically the phase and gain response of $h[n]$. **[4 Marks]**

c) Filter specifications for low-pass digital and analog filters are shown below, respectively.



Using the Bilinear Transformation, the goal is to design a discrete-time IIR filter that satisfies the frequency response constraints. Furthermore, it is desirable that the peak gain (digital filter) in the passband is unity, that is,

$$\max_{\omega\in[0,\omega_p]} |H(e^{\jmath\omega})| = 1.$$

Given that for a digital filter, the filter order is estimated by,

$$M \approx -\frac{1}{|\omega_s - \omega_p|}\left(\frac{56}{10} + \frac{43}{10}\log_{10}\left(\varepsilon_s \varepsilon\right)\right),$$

compute the filter order in terms of analog filter parameters μ and λ. **[4 Marks]**

d) Let $H\left(\jmath\omega\right)$ be an all-pass filter with a non-linear phase response $\angle H\left(\jmath\omega\right) = \beta\omega^3 - \frac{\omega^2}{2}$. Let $H_1(j\omega)$ be a filter with all-pass gain and phase given by

$$\angle H_1\left(\jmath\omega\right) = \angle H\left(\jmath\omega_0\right) + \frac{\left(\omega - \omega_0\right)^2}{2}\frac{d^2\angle H}{d\omega^2}\left(\jmath\omega_0\right). \tag{5}$$

i) Compute the frequency response $H_1\left(\jmath\omega\right)$. **[2 Marks]**

ii) Let $H_2\left(\jmath\omega\right)$ be an all-pass filter with phase

$$\angle H_2\left(\jmath\omega\right) = \angle H\left(\jmath\omega_0\right) + \frac{\left(\omega - \omega_0\right)^2}{2}\frac{d^2\angle H}{d\omega^2}\left(\jmath\omega_0\right) + \frac{\left(\omega - \omega_0\right)^3}{6}\frac{d^3\angle H}{d\omega^3}\left(\jmath\omega_0\right). \tag{6}$$

For $\omega_0 = 0$, compute $H_2\left(\jmath\omega\right) - H\left(\jmath\omega\right)$. **[3 Marks]**

iii) Find β as a function of ω_0 such that $\omega_0 \cdot \tau_H\left(\jmath\omega_0\right) = \angle H_1\left(\jmath\omega_0\right)$ **[3 Marks]**

Datasheet:

Standard Sequences

- $\delta[n] = 1$ for $n = 0$ and 0 otherwise.
- $\delta_{\text{condition}}[n] = 1$ whenever "condition" is true and 0 otherwise.
- $u[n] = 1$ for $n \geq 0$ and 0 otherwise.

Geometric Progression

- $\sum_{n=0}^{r} \alpha^n z^{-n} = \frac{1-\alpha^{r+1} z^{-r-1}}{1-\alpha z^{-1}}$ provided that $\alpha z^{-1} \neq 1$.
- $\sum_{n=0}^{\infty} \alpha^n z^{-n} = \frac{1}{1-\alpha z^{-1}}$ provided that $\left|\alpha z^{-1}\right| < 1$.

Forward and Inverse Transforms

z:	$X(z) = \sum_{-\infty}^{\infty} x[n] z^{-n}$	$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$
CTFT:	$X(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega$
DTFT:	$X(e^{j\omega}) = \sum_{-\infty}^{\infty} x[n] e^{-j\omega n}$	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$
DFT:	$X[k] = \sum_{0}^{N-1} x[n] e^{-j2\pi \frac{kn}{N}}$	$x[n] = \frac{1}{N} \sum_{0}^{N-1} X[k] e^{j2\pi \frac{kn}{N}}$
DCT:	$X[k] = \sum_{n=0}^{N-1} x[n] \cos \frac{2\pi(2n+1)k}{4N}$	$x[n] = \frac{X[0]}{N} + \frac{2}{N} \sum_{n=1}^{N-1} X[k] \cos \frac{2\pi(2n+1)k}{4N}$
MDCT:	$X[k] = \sum_{n=0}^{2N-1} x[n] \cos \frac{2\pi(2n+1+N)(2k+1)}{8N}$	$y[n] = \frac{1}{N} \sum_{0}^{N-1} X[k] \cos \frac{2\pi(2n+1+N)(2k+1)}{8N}$

Convolution

DTFT: $v[n] = x[n] * y[n] \triangleq \sum_{r=-\infty}^{\infty} x[r] y[n-r] \quad \Leftrightarrow \quad V\left(e^{j\omega}\right) = X\left(e^{j\omega}\right) Y\left(e^{j\omega}\right)$

$v[n] = x[n] y[n] \quad \Leftrightarrow \quad V\left(e^{j\omega}\right) = \frac{1}{2\pi} X\left(e^{j\omega}\right) \circledast Y\left(e^{j\omega}\right) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X\left(e^{j\theta}\right) Y\left(e^{j(\omega-\theta)}\right) d\theta$

DFT: $v[n] = x[n] \circledast_N y[n] \triangleq \sum_{r=0}^{N-1} x[r] y[(n-r)_{\mod N}] \quad \Leftrightarrow \quad V[k] = X[k] Y[k]$

$v[n] = x[n] y[n] \quad \Leftrightarrow \quad V[k] = \frac{1}{N} X[k] \circledast_N Y[k] \triangleq \frac{1}{N} \sum_{r=0}^{N-1} X[r] Y[(k-r)_{\mod N}]$

Group Delay

The group delay of a filter, $H(z)$, is $\tau_H(e^{j\omega}) = -\frac{d\angle H(e^{j\omega})}{d\omega} = \Re\left(\frac{-z}{H(z)} \frac{dH(z)}{dz}\right)\Big|_{z=e^{j\omega}} = \Re\left(\frac{\mathscr{F}(nh[n])}{\mathscr{F}(h[n])}\right)$ where $\mathscr{F}(\,)$ denotes the DTFT.

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Order Estimation for FIR Filters

Three increasingly sophisticated formulae for estimating the minimum order of an FIR filter with unity gain passbands:

1. $M \approx \frac{a}{3.5\Delta\omega}$
2. $M \approx \frac{a-8}{2.2\Delta\omega}$
3. $M \approx \frac{a-1.2-20\log_{10} b}{4.6\Delta\omega}$

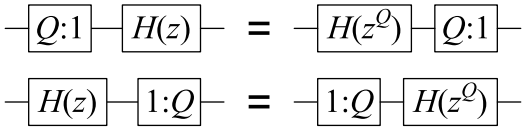
where a =stop band attenuation in dB, b = peak-to-peak passband ripple in dB and $\Delta\omega$ = width of smallest transition band in radians per sample.

z-plane Transformations

A lowpass filter, $H(z)$, with cutoff frequency ω_0 may be transformed into the filter $H(\hat{z})$ as follows:

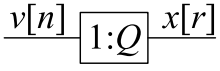
Target $H(\hat{z})$	Substitute	Parameters
Lowpass $\hat{\omega} < \hat{\omega}_1$	$z^{-1} = \frac{\hat{z}^{-1}-\lambda}{1-\lambda\hat{z}^{-1}}$	$\lambda = \frac{\sin\left(\frac{\omega_0-\hat{\omega}_1}{2}\right)}{\sin\left(\frac{\omega_0+\hat{\omega}_1}{2}\right)}$
Highpass $\hat{\omega} > \hat{\omega}_1$	$z^{-1} = -\frac{\hat{z}^{-1}+\lambda}{1+\lambda\hat{z}^{-1}}$	$\lambda = \frac{\cos\left(\frac{\omega_0+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\omega_0-\hat{\omega}_1}{2}\right)}$
Bandpass $\hat{\omega}_1 < \hat{\omega} < \hat{\omega}_2$	$z^{-1} = -\frac{(\rho-1)-2\lambda\rho\hat{z}^{-1}+(\rho+1)\hat{z}^{-2}}{(\rho+1)-2\lambda\rho\hat{z}^{-1}+(\rho-1)\hat{z}^{-2}}$	$\lambda = \frac{\cos\left(\frac{\hat{\omega}_2+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)}$, $\rho = \cot\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)\tan\left(\frac{\omega_0}{2}\right)$
Bandstop $\hat{\omega}_1 \not< \hat{\omega} \not< \hat{\omega}_2$	$z^{-1} = \frac{(1-\rho)-2\lambda\hat{z}^{-1}+(\rho+1)\hat{z}^{-2}}{(\rho+1)-2\lambda\hat{z}^{-1}+(1-\rho)\hat{z}^{-2}}$	$\lambda = \frac{\cos\left(\frac{\hat{\omega}_2+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)}$, $\rho = \tan\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)\tan\left(\frac{\omega_0}{2}\right)$

Noble Identities



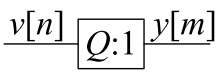
Multirate Spectra

Upsample:



$$x[r] = \begin{cases} v\left[\frac{r}{Q}\right] & \text{if } Q \mid r \\ 0 & \text{if } Q \nmid r \end{cases} \quad \Rightarrow \quad X(z) = V(z^Q)$$

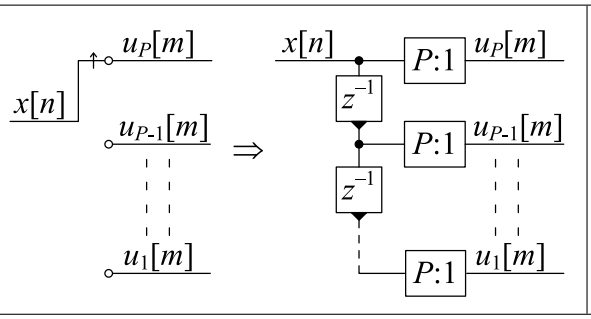
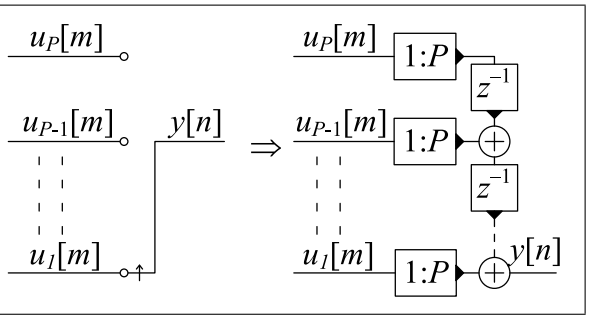
Downsample:



$$y[m] = v[Qm] \quad \Rightarrow \quad Y(z) = \frac{1}{Q}\sum_{k=0}^{Q-1} V\left(e^{\frac{-j2\pi k}{Q}} z^{\frac{1}{Q}}\right)$$

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Multirate Commutators

Input Commutator	Output Commutator
	

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